FORMALIZATION OF THE SOCIAL LOAFING THEORY

Practical Course in Empirical Research Methods

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1. Objektive

The current psychological theories mainly consist of verbal theories. These are mostly vague and not specific. Often additional assumptions are needed, and falsifications can therefore not be attributed to the theory itself but only the implementations (van Dongen et al., 2024). To address this issue the present work aims to formalize the social loafing theory. Specifically, a formal model is developed and evaluated using simulated data.

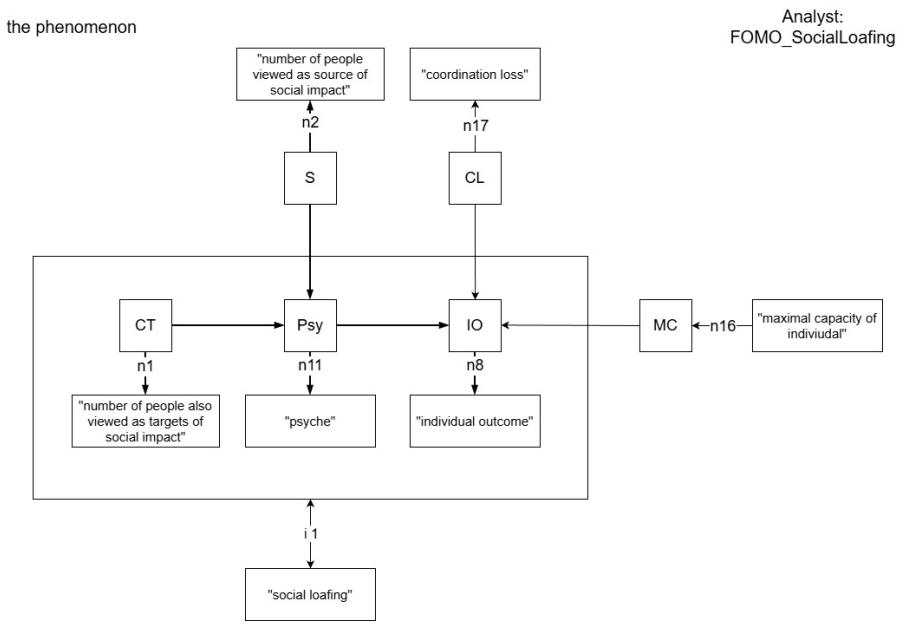
2. Verbal introduction

The social loafing theory suggests that "people exhibit a sizable decrease in individual effort when performing in groups as compared to when they perform alone" (Latané et al., 1979). This theory was selected because, despite extensive empirical research, the relationships between its core constructs remain underspecified. A formalization may therefore provide a clearer basis for further research.

3. Phenomenon

Figure 1

VAST of the phenomenon



As Figure 1 shows, the central phenomenon is that an increasing number of co-targets (CT) reduces individual output (IO). The number of sources (S) have the contrary effect.

An increasing number of sources (S) should increase the individual output (IO). Additionally, the individual output (IO) depends on the personal maximal capacity (MC) as well as the coordination loss (CL) that occurs when several people act simultaneously.

The meta-analysis of Karau and Williams (1993) extensively studied this phenomenon. The studies they resorted to did neither include a variation of S nor an examination of CL. Therefore, they only verified a part of the presented phenomenon with additional variations. Nevertheless, the results suggest a robust phenomenon regarding the link between an increasing number of CT and a decrease in IO. They examined 78 studies with almost 8000 participants. The meta-analysis reported a significant mean effect size ($d = 0.44$, $p < .05$), suggesting that the phenomenon of social loafing is significant.

The UTOS framework (Cronbach & Shapiro, 1982) is used to assess generalizability. Firstly, the meta-analysis suggests that the phenomenon is universal. The participants (units) differed regarding culture, gender, and age. Despite these differences, the effect of social loafing was observed across populations. But the exact effect size between populations varied a lot. The medium effect size could mostly be shown for western men. Women also exhibited social loafing, albeit to a lesser extent. Similarly, smaller effects were observed in Eastern cultures. Regarding the age, a heterogeneity in the exact effect size can be observed. Overall, social loafing can be observed in individuals of all ages. The included experiments mostly studied college students. The heterogeneity can therefore also be explained due to the lack of data. In conclusion, the effect of social loafing is universal whereas the effect size is not.

Secondly, the meta-analysis examined several treatments. The task complexity as well as the type of effort and tasks have been varied. Physical, cognitive, perceptual and evaluative efforts have been measured as well as maximizing, optimizing, additive and compensatory tasks. The type of effort and task did not impact the effect size of social loafing. The task complexity on the other hand had a huge impact. Social loafing could only be found for simple tasks but not for complex tasks.

Thirdly, due to different treatments outcomes were also measured differently. They did not report people conducting the same experiment and only varied the tasks. Therefore, outcome generalizability can be suspected due to the task variety but not proven.

Lastly the experiments were conducted in different settings. The phenomenon could be found in the laboratory as well as in the field. But anew, the effect size varied. In the laboratory condition social loafing is more prominent then in the field.

In conclusion, the phenomenon can be considered robust because it is supported by a substantial body of evidence (d = 0.44) and generalizes across units, treatments, outcomes, and settings, although effect sizes vary across conditions. Additionally, the findings limit the scope. Social loafing only occurs in simple tasks.

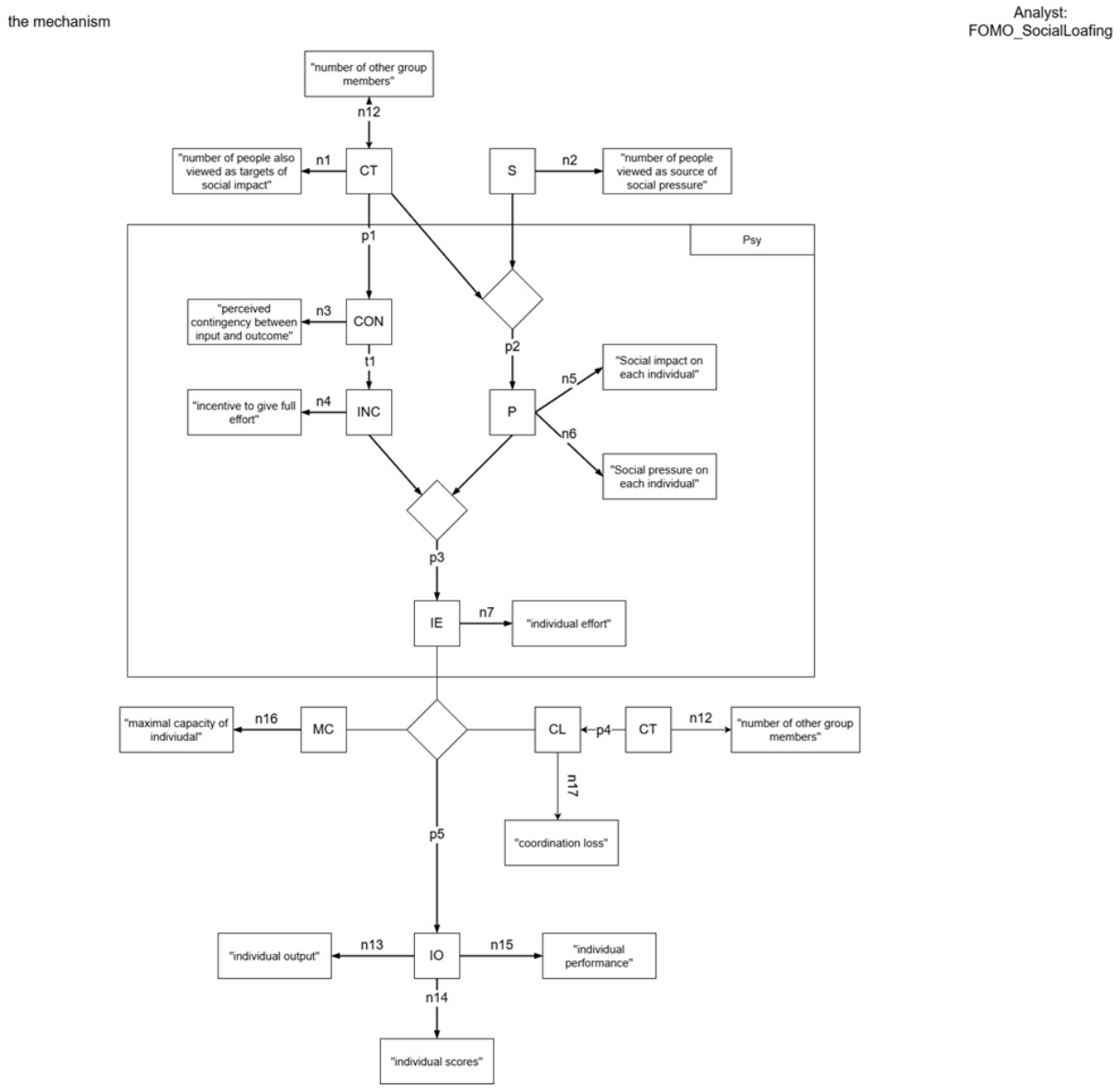
4. Core Construct and Formalization of a (Proto)theory

In the following section, the core constructs will shortly be described and how they deviate from the original literature as well as the links between them and how the theory accounts for the phenomenon. A summary of all construct definitions and their sources (Latané et al., 1979) is provided in Table A1 in the appendix.

The focus of the social loafing theory is that the more people are in a group the less every single person performs in contrast to performing alone. Thereby, the number of co-targets (CT) in a group considerably influences the individual output (IO) of every single person. Hereby, the CT are defined as the number of other people in a group without counting oneself and the IO is the individual output a person produces on average. Latané et al. (1979, p. 829) also called IO performance. The average outcome has been used because the IO could be affected by the other participants due to interfering sound pressure waves. The paper of Latané et al. (1979) suggests two ways the CT influence individual effort (IE) a person shows. Firstly, the number of CT impacts the perceived contingency (CON) between what a person gives to a group in form of effort and what they receive in return. The larger the number of CT the less one is affected by the praise and punishment regarding one's own effort. This leads to a decreased perceived CON. This perceived contingency directly relates to the incentive (INC) a person has, to give their full effort. Secondly, the number of CT influences how much social pressure (P) a single person feels. The more people are in a group, the more the pressure is equally distributed amongst group members. Every single person therefore feels less P. Latané et al. (1979, p. 825) also used the term of social impact for P. The number of sources (S) – being people who observe the group like experimenters – also impact P. A higher number of S increases P. The individual effort (IE) a single person makes is based on a combination of P and INC.

Figure 2

VAST of the mechanism



Because effort alone can hardly be measured, the individual output (IO) is needed to compare groups. The IO is dependent on the maximal capacity (MC) a person has and the coordination loss (CL) occurring when many people perform a task simultaneously. The MC has not been defined by Latané et al. (1979) but is needed to formalize the verbal theory. The CL in this work is defined by the three categories Latané et al. (1979, p. 826) suggested to be inherent to CL. These three factors are sound cancellation due to interfering sound pressure waves, directional coordination loss due to voices being projected towards various locations, and temporal coordination loss due to different timing of the intensity of the voices. This exact definition limits the scope of the theory

because every task has its own kind of CL. This definition is only viable for tasks involving sound. Other papers also suggest other definitions. Miles et al. (2017) for example proposes a definition that is more socially oriented. This definition would miss the physical aspect that is inherent to a quantity of tasks. The definition of CT and S Latané et al. (1979, p. 823) have given has been simplified. Instead of examining the effect of the strength, immediacy and number of the CT and S we focused only on the number of the CT and S. Other than that, we did not deviate from the original definitions. As can be seen in the VAST (see Figure 2), the authors used some "jangle" (Kelly, 1927). They used different words to describe the same construct. We noted these inconsistencies in the VAST. In the following work I will be using one phrase for every construct to avoid confusion.

5. Development of formal model

To develop a formal model, we first created a Visual Argument Structure Tool (VAST) (Leising et al., 2023). VAST is a short, graphic visualization of the theory that shows the constructs as well as the relationships between them. We created it with the application draw.io. To simulate the data, we used R. All materials and code are available at https://github.com/LeonieFeltesse/SLoafing_final.

For the formalization we closely followed the suggestions of Latané et al. (1979). Unfortunately, his explanations remained mostly vague. Thus, several assumptions had to be introduced. For instance, he explained the effect of S in the beginning of his paper but later only specified the effect of CT. He developed a function for this relationship without integrating S and never varied S in his experiments. Therefore, a plausible effect of S had to be assumed. On another occasion, he mixed up contingency and incentive. Staying true to the definition of each, we had to separate both variables in our model. We had to develop the relationship with common sense instead of solely on the paper. We did not willingly change his theory. The only thing we narrowed down were the definition of CT and S. He suggested that they have three dimensions: immediacy, strength and number. He did not define immediacy and strength any further and all three dimensions would have been too complex for our model. Therefore, we only focused on the number of CT and S.

The variables mentioned above influence the individual effort (IE). But to transfer IE into an output we need to integrate the maximal capacity. This was a necessity for the formalization but not derived from the theory of Latané et al. (1979). Another construct whose effect Latané et al. (1979) didn't specify is CL. As mentioned above CL has many reasons and occurs when people perform a task simultaneously. Depending on the number of people, the task as well as the environment, its effect varies a lot. CL is the reason why the group output is less than every single individual output combined, even if every single person would give the same effort. Due to its complexity, I only defined it in dependency of the number of CT. With increasing numbers of CT, CL also increases. But, literature suggests, that this increase is not infinite and finally reaches a plateau (El Asri et al., 2023). To define the slope, I designed the graph after the coordination loss visible in the results of Latané et al. (1979) comparing actual groups (with CL) with pseudo groups (without CL).

The exact deviations of the constructs and their relationships are provided in Appendix A (Table A2).

The variables are defined in Table 1.

Table 1

Variable Table:

Construct	Source (if given)	Scale level	Range/Values	Anchors
CT	"and *number* of group members" (Latané et al., 1979, p. 823)	discrete	$\in \mathbb{N}_0$	0 = "zero cotargets are present"; n = "number of cotargets (other persons in the evaluated group, number of group members minus 1)"
S	"*number* of these other persons." (Latané et al., 1979, p. 823) AND "Thus, as the *number of performers decreased,*	discrete	$\in \mathbb{N}_0$	"0 = "zero persons that cause the social pressure"; n = "number of persons that cause the pressure"

	the size of the audience increased. According to Cottrell's evaluation apprehension hypothesis (1972), the presence of an evaluative audience should enhance performance for a simple, well learned task such as rope pulling, and, although there is little supportive evidence, it seems reasonable that the *larger the audience*, the greater the enhancement“ (Latané et al., 1979, p. 826)			
P		continuous	∈ [0; 1[	0 = “zero persons are source of social pressure”, “no social pressure” 1 = “maximal social pressure (a theoretical amount if infinite sources and zero cotargets are present)”
CON	“Individuals could "hide in the crowd" (Davis, 1969) and avoid the negative consequences of slacking off, or they may have felt "lost in the crowd" and unable to obtain their fair share of the positive consequences for working hard.” (Latané et. al, 1979, p. 830) “reward” (Latané et. al, 1979, p. 830)	continuous	∈ [0; 1]	0 = “expectation that the share (reward/punishment) of each member is completely independent of what they’ve given” 1 = “expectation that their share (reward/punishment) is equal to what they’ve given”
INC		continuous	∈ [0; 1]	0 = “no incentive to give any individual effort” ;

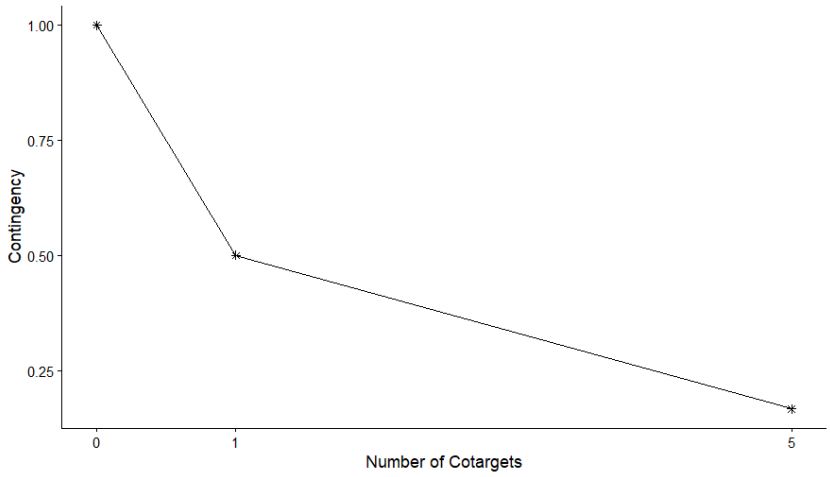
				1 = “incentive to give full effort”
IE	e. g. “dyads pulled at 91% of the sum of their individual capacities, trios at 82%, and groups of six at only 78%” (Latané, 1979, p. 825)	continuous	∈ [0; 1]	0 = “the person shows no effort at all to bring in some contribution to the group task”, 1 = “the person shows their maximal effort in contributing to the group task”
IO	“dyads pulled at 91% of the sum of their individual capacities, trios at 82%, and groups of six at only 78%” (Latané, 1979, S. 825)	continuous	∈ [0; inf[	depends on task; in Latane et al.’s study (1979) 0 = 0 dynes/cm^2 n = n dynes/cm^2
MC		continuous	∈ [0; inf[	depends on task; in Latane et al.’s study (1979) 0 = 0 dynes/cm^2 n = n dynes/cm^2
CL	„ (a) Sound cancellation will occur to the extent that sound pressure waves interfere with each other, (b) directional coordination losses will occur to the extent that voices are projected toward different locations, and (c) temporal coordination losses will occur to the extent that moment-to-moment individual variations in intensity are not in synchrony.” (Latané, 1979, p. 826)	continuous	∈ [0; 4[	depends on task; in Latane et al.’s study (1979) 0 = no coordination loss 4 = the IO is reduced by 4 dynes/cm^2/ maximal coordination loss possible/ infinite numbers of cotargets

Additionally, each relationship is illustrated by a plot accompanied by a brief textual explanation.

Figure 3 shows that CON is dependent on CT. As defined by Latané et al. (1979) this relationship follows the principle of 1/CT+1. Every additional CT has less influence than the CT before. With increasing number of CT, the CON diminishes.

Figure 3

Contingency between input and outcome dependent on CT



As illustrated in Figure 4 the perceived CON between input and output can be directly transformed into INC:

Figure 4

Incentive to give full effort dependent on CON

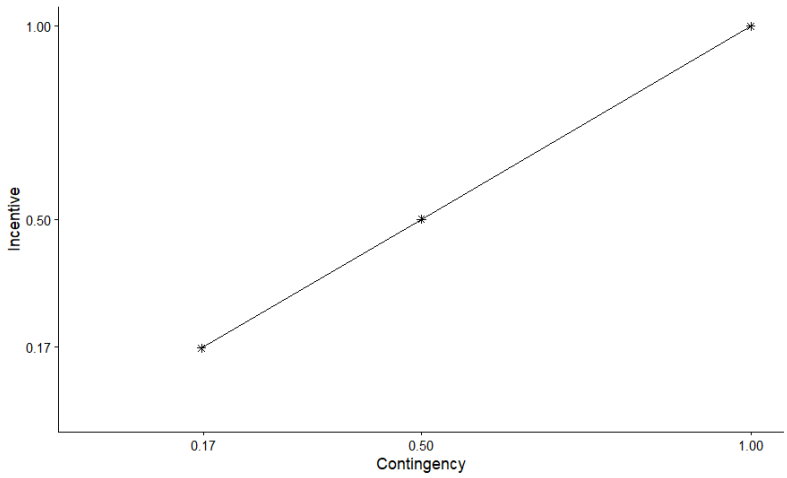


Figure 5 depicts the effect of CT and S on P. An increasing number of CT decreases P. S on the other hand increase the social pressure. Latané et al. (1979) did not specify which construct (CT or S) had the greater influence on the social pressure.

Figure 5

Pressure dependent on CT and S

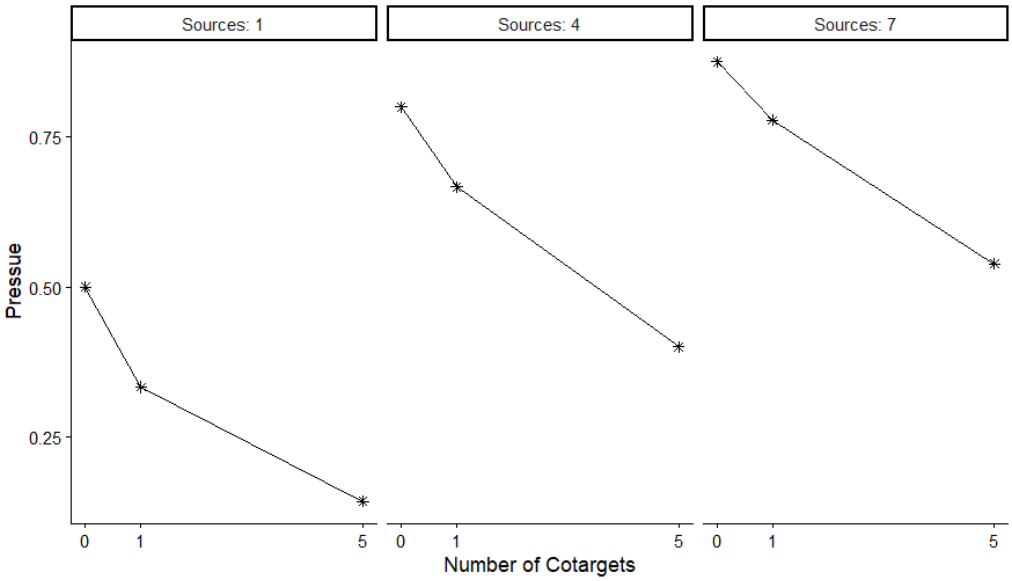
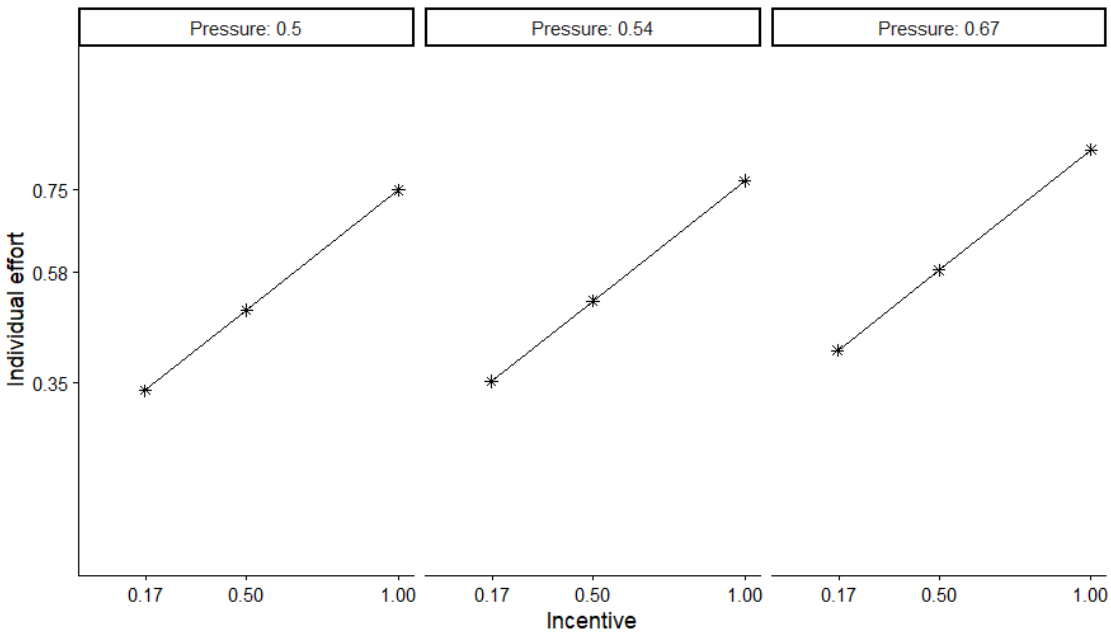


Figure 6 displays that IE is dependent on INC and P. The higher INC and P, the higher IE. Anew, Latané et al. (1979) did not specify which variable (INC or P) had the greater influence. Therefore, we assumed that INC and P influenced IE equally.

Figure 6

Indiviual effort dependent on P and INC



As Figure 7 shows, an increasing number of CT also increases CL. Every additional person first has a great impact and then an increasingly smaller one.

Figure 7

Coordination loss dependent on CT

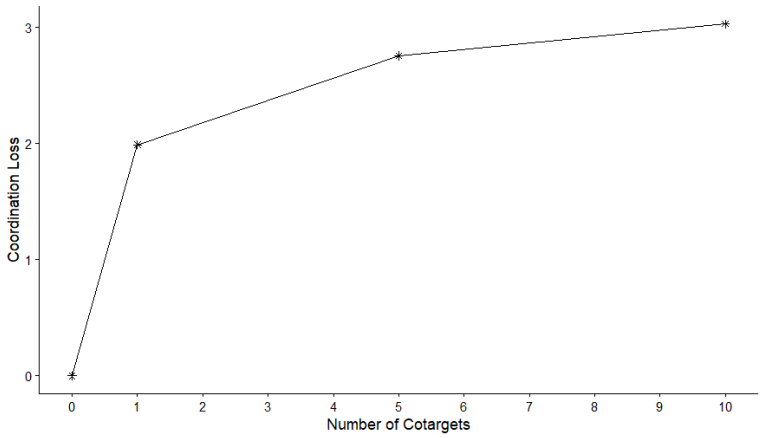


Figure 8 illustrates that IO increases with IE depending on the MC. Additionally, CL diminishes IO.

Figure 8

Individual outcome dependent on MC, IE and CL

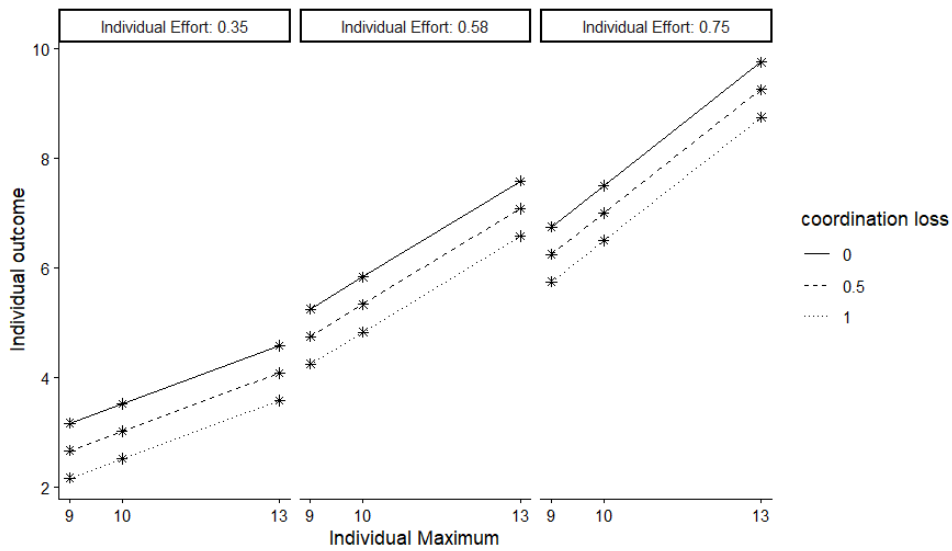
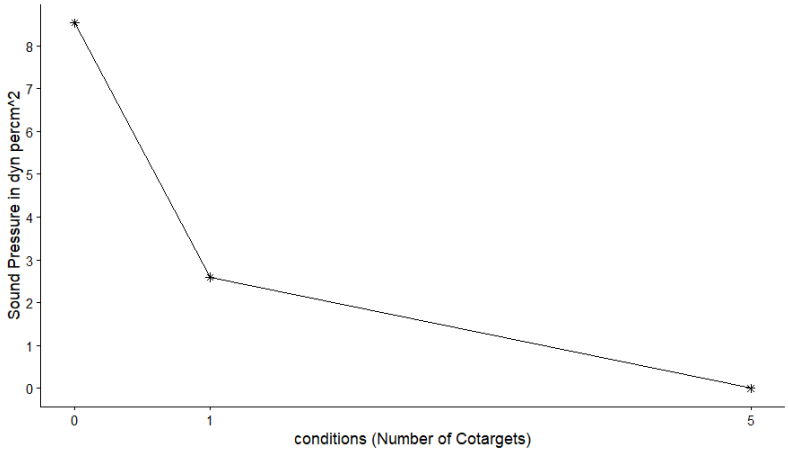


Figure 9 presents how simulated data could look. Due to social loafing and CL (whilst keeping the number of S consistent) every additional CT diminishes IO.

Figure 9

Mean individual outcome dependent on experimental group (while S=1)



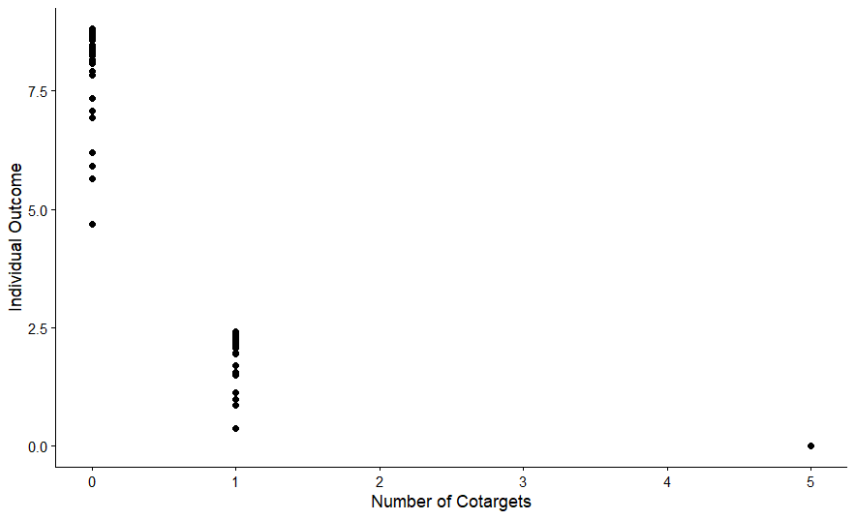
6. Evaluation of the Formal model

Following the Experiment 1 of Latané et al. (1979), an experiment with 6 groups of 6 people and one experimenter (=1 source) per group was simulated. Therefore, the experiment consisted of 36 participants. Every person had to cheer and clap in three conditions: alone, with one CT and with 5 CT. S was not varied.

The scatterplot in Figure 10 displays that the individual outcome did decrease with the number of CT.

Figure 10

Individual output dependent on CT



A one-way ANOVA (see table 1) was conducted to examine the effect of the number of CT on IO. The analysis revealed a significant effect of group size, $F(2, 645) = 4969$, $p < .05$. The observed effect size was very large ($\eta^2 = .94$), indicating that the number of CT explains most of the variance in individual outcomes.

Table 1

ANOVA Results for the Effect of CT on IO

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
CT_factor	2	8277	4139	4969	<2e-16 ***
Residual	645	537	1		

To examine whether this effect can be attributed solely to CL, the analysis (see Table 2) was repeated using the individual output without CL. Even without CL the analysis revealed a significant effect of group size, $F(2, 645) = 2502$, $p < .05$. The effect of group size remained very large ($\eta^2 = .89$), suggesting that the reduction in individual outcome cannot be explained by coordination loss alone but reflects social loafing.

Table 2

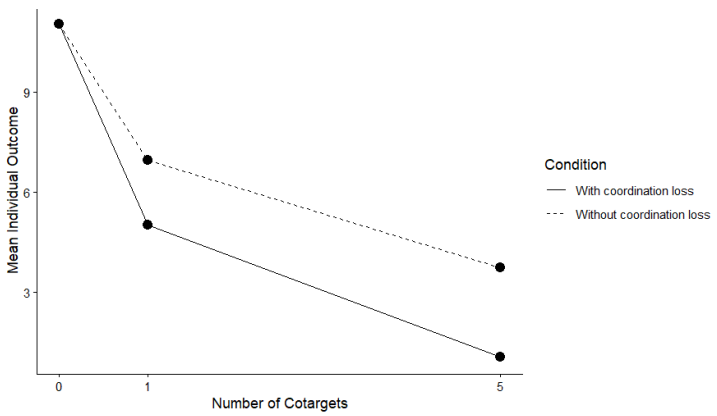
ANOVA Results for the Effect of CT on IO without CL

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
CT_factor	2	5457	2728.1	2502	<2e-16 ***
Residual	645	703	1.1		

Figure 11 visualizes this outcome in form of a graph comparing IO with and without coordination loss.

Figure 11

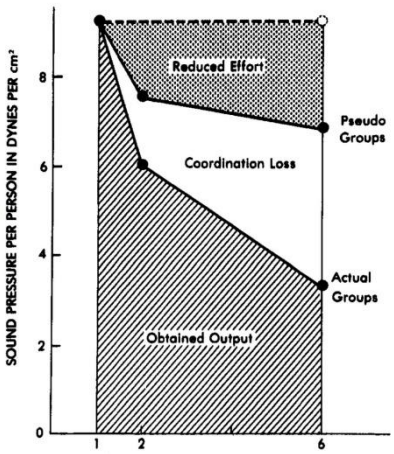
Comparison of individual outcomes with and without CL



These results look similar to the comparison between the two experiments of Latané et al. (1979) (see Figure 12) between Actual Groups with CL (called Actual Groups in the figure) and Pseudo Groups without CL.

Figure 12

Comparison sound pressure between Pseudo Groups and Actual Groups



Note. Data shown for comparison; original from Latané, Williams, & Harkins (1979, Figure 2).

The decrease of the individual output due to social loafing and the coordination loss is higher in the simulated data than in the study of Latané et al. (1979). Nevertheless, the effect could be found. As mentioned in the meta-analysis of Karau and Williams (1993) the effect of social loafing varies a lot across conditions and populations. That's why the findings of this work – although it being a larger effect size then what has been found before - supports the existence of the social loafing effect, especially the effect of the CT. Nevertheless, this study shows that the verbal theory lacks precision, since many relationships are subject to individual interpretation. More research and specification concerning the exact mechanism behind social loafing is needed.

7. Meta-Reflection

Formalization was challenging due to underspecification in the original theory. For example, the standard deviation from the results in the paper of Latané et al. (1979) have not been reported. Therefore, it was necessary to freely choose the amount of randomness incorporated into the formalization. A value of 0.1 was chosen to reflect randomness arising

from fatigue, personality, and motivation. Furthermore, the effect of S has not been clarified, leaving gaps to close. Another difficulty has been to formalize the coordination loss. According to the results it should make up 3 to 4 dynes per cm^2 in the condition with 5 CT. Due to the social loafing effect itself already diminishing the individual outcome by a lot, the coordination loss leaded the individual outcome to go to 0. Meaning that if 6 people simultaneously cheer and clap the pressure of the sound waves overlap in a way that takes away every sound. A scenario that is rather unrealistic. To solve this problem, we would have needed to increase the effect of the sources, an adjustment we had no evidence for. Additionally, the authors used a lot of "jangle" (Kelly, 1927, p.64), using different words for the same thing. For instance, they used social pressure and social impact interchangeably, as well as individual effort and performance or contingency and incentive, causing confusion. To conclude, the original literature has been helpful to a certain degree but integrating own thoughts was highly needed. This shows that verbal theories often are not specific enough and need a specification, for example in form of a formalization ideally proposed by the authors themselves.

Besides the vague nature of the theory the formalization has been hard due to personal skills. Before this seminar I was not very versed with coding, making the formalization in R rather difficult. The seminar was very well structured which helped with this issue.

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Appendix A

Table A1:

Construct Source Table

ID	**Type** - **P**henomenon - **C**oncept - **HOC**	**Short name**	**Quote**	**Reference**	**Rel. type (n, p, i, r, ...)**	**Comment (e.g., identified gaps & inconsistencies)**	**Incl. (Y/N)?**
CT	C	co-targets	- "According to Latane's (1981) social impact theory, people can be viewed as [...] targets of social impact." (p. 682) AND - "and number of group members" (Latané et al., 1979, S. 823) - "impact is divided among the group	Latané et al., p.823 (1979) AND Karau & Williams, p. 682 (1993)	n1, p1, p2, n12		Y

			members ” (Latané et al., 1979, p. 830) - “However, if a person is a member of a group that is the target of social forces from outside the group” (Latané et al., 1979, p. 823)				
S	C	Source of pressure	- “The impact that the experimenters have on an individual […]” (p.825) - “According to Latane's (1981) social impact theory, people can be viewed as either sources	Latané et al., S.825 (1979) AND Karau & Williams, S. 682 (1993)	n2, p2	these persons are not described as a pressure precisely, but we can assume that they are the reason why the pressure occurs pressure = impact	Y

			[...] of social impact." (p. 682)				
CON	C	Contingency between input and outcome	"[...] the contingency between their input and the outcome [...]" (p. 830)	Latané et al., p. 830 (1979)	p1, n3, t1	here, contingency is simply explained through the word itself, no further definition	Y
INC	C	Incentive	"[...] incentive to work equally hard in groups of different sizes." (p.830)	Latané et al., p. 830 (1979)	t1, n4, p3		Y
P	C	impact/ pressure	"the amount of social pressure on the target person" (p. 823) "pressures on each individual" (p. 830) / "In a group performance situation in which pressures	Latané et al., p.823 (1979)	p2, n5, n6, p3	impact and pressure are no really distinctive concepts, cause impact can officiate as a social facilitation. here, we assume its the same cause we limited the scope to the negative form of social impact, no facilitation at all	Y

			to work come from outside the group" (p. 830) AND "[…]the impact of these forces on any given member […]" (p. 823)				
IE	C	individual effort	"individual effort" (p. 822)	Latané et al., p.822 (1979)	p3, n7, p5	In this paper, there is no real distinction between "effort" and "performance". These concepts were used partially synonymously. The problem is, they were not defined.	Y
Psy	HOC	Psyche	no citation	/	n11	We defined this HOC as the space, in which the mental process is defined	Y

IO	C	individual output	"in this case n times the average individual output" (p. 825) AND "Since individual scores" (p. 830) AND - "leading to an apparent drop in individual performance, a phenomenon we call social loafing" (p. 825) - "sum of their individual performances" (p. 829)	Latané et al., p.825/p.829/p.830 (1979)	n13, n14, n15		
MC	C	maximal capacity	"dyads pulled at 91% of the sum of their individual **capacities**, trios at 82%, and groups of	Latané et al., p.825 (1979)	n16, p5	maximal capacity of individual, when 100 sources and 0 cotargets are present	Y

			six at only 78%” (p. 825)			(no citations are available, this is our own interpretation to simulate the phenomenon)	
CL	C	coordination loss	„As group size increases, the number of "coordination links," and thus the possibility of faulty coordination (pulling in different directions at different times), also increases. “	Latané et al., S. 825 (1979)	p3, p5, n17		Y

Table A2

Relationship table

Output variable	**Function to compute the output variable**	**Comment**	**To what extent derived from theory?[a]**
P1	P1= 1/n		“[…]the amount of

			social pressure on the target person should increase as a multiplicative function of the strength, immediacy, and number of these other persons.” (Latané, 1979, p.823) => derived from theory CT → CON: “We believe that social loafing occurs because there is usually a stronger perceived contingency between individual effort and valued outcomes when working individually. When working collectively, factors other than the individual's effort frequently determine performance, and valued outcomes are often **divided among all of the group members**.“ Karau & Williams, p. 684 (1993) “which both suggest that individuals' outcomes are more contingent on their efforts when working with others who are expected to perform poorly.“ Karau & Williams, p. 696 (1993)
P2	P2 = S/(S+CT+1)		“Impact is divided up among the group members, in much the same way that responsibility for helping seems to be divided among witnesses to an emergency” (Latané, 1979, p.823) => derived from theory

			“If a person is the target of social forces, increasing the number of other persons also in the target group should diminish the pressures on each individual because the impact is divided among the group members.” (Latané, 1979, p. 830) CT → P: “The more people who are the target of this pressure, the less pressure should be felt by any one person.” (Latané, 1979, p. 823) => derived from theory “increased group size to result in reduced effort” (Latané, 1979, p. 823)
P3	P3 = 0.5 * INC + 0.5 * P	Additional randomness with sd = 0,1 due to personality, fatigue, motivation,...	“the nth person should have less effect than the (n - l)th.” (Latané, 1979, p. 823) “Just as perceptual judgments of physical stimuli follow power functions (Stevens, 1957), so also should judgments of social stimuli, and the exponent of the psycho social power function should have an exponent of less than one, resulting in a marginally decreasing impact of additional people.“ (Latané, 1979, p. 823, 830) P → IE: “The more people who are the target of this pressure, the less pressure should be felt by any one person.

			Since people are likely to work hard in proportion to the pressure they feel to do so, we should expect increased group size to result in reduced efforts on the part of individual group members.” (Latané, 1979, p. 823) “If a person is the target of social forces, increasing the number of other persons also in the target group should diminish the pressures on each individual because the impact is divided among the group members. In a group performance situation in which pressures to work come from outside the group and individual outputs are not identifiable, this division of impact should lead each individual to work less hard. Thus, whether the subject is dividing up the amount of work he thinks should be performed or whether he is dividing up the amount of reward he expects to earn with his work, he should work less hard in groups.” (Latané, 1979, p. 830)
P4	P4 = 4*sqrt(CT)/(1+sqrt(CT))	Additional randomness with sd = 0,05 due to room size, indoor vs outdoor, furniture, distance between participants,...	Possible reasons for coordination loss: „ (a) Sound cancellation will occur to the extent that sound pressure waves interfere with each other, (b) directional coordination losses will occur to the extent that voices are

			projected toward different locations, and (c) temporal coordination losses will occur to the extent that moment-to-moment individual variations in intensity are not in synchrony." (Latané et al., 1979, p. 826) =>therefore the more people, the more coordination loss =>0 CT sound waves/... should not interfere, not coordination loss => the exact slope of the coordination loss is not derived from theory
t1			=> Not based on focal theory, but rather on common sense or other theories "Thus, under these assumptions, if his own performance cannot be individually monitored, an individual's incentive to perform should be proportional to l/n." (Latané et al., 1979, p. 830)
P5	P5 = IE*MC - CL		=> Not based on focal theory, but rather on common sense or other theories